

# MIGUEL BLANCO GARCÍA

SCIENTIFIC PROJECT COORDINATION · MATERIALS · SEMICONDUCTORS

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## PROFILE

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Materials scientist, PhD physicist and chemist by training, combining scientific depth with project coordination, experimental delivery and applied AI. Experienced in turning complex technical questions into structured programmes across international research facilities, multidisciplinary teams and industrial R&D.

## PROFESSIONAL EXPERIENCE

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2022—2026

### PhD Researcher in Physics

[DESY NanoLab / University of Hamburg](#)

Led a scientific work package in the EU-funded CORAERO project. Designed experimental programmes for photocatalytic semiconductors, coordinated measurements and collaborators, and delivered synchrotron campaigns at DESY, SOLEIL and ESRF. Operated and supported STM/SPM, XPS and FT-IRRAS systems. Produced four peer-reviewed ACS publications, including two first-author JACS papers.

2021—2022

### R&D Plastics Intern

[Repsol](#)

Supported the RECICLEX project on recycled polymers and polyolefin formulations. Evaluated rheological, thermal and mechanical properties, investigated anomalies and prepared documentation for R&D, quality and industrial teams.

2020—2021

### Research Trainee

[Universidad Autónoma de Madrid](#)

Studied magnesium ultrathin films for hydrogen storage and photo-assisted hydrogen production. Prepared electrochemical cells and performed cyclic voltammetry, impedance measurements and data analysis.

## EDUCATION

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2026

### PhD in Physics — summa cum laude

[University of Hamburg](#)

Surface science, semiconductor materials, thin films, synchrotron experiments and light-driven reaction mechanisms.

## EDUCATION CONTINUED

- 2021** **MSc in Renewable Energy Materials — 9.00/10**  
**Universidad Autónoma de Madrid**  
Hydrogen, electrochemistry, photovoltaics and sustainable energy systems. Master's thesis on magnesium ultrathin films: 9.2/10.
- 2020** **BSc in Chemistry — 8.44/10**  
**Universidad Autónoma de Madrid**  
Ranked 10th among 138 graduates that year. Broad chemistry and electrochemistry training.
- 2018—2019** **Erasmus+ in Chemistry**  
**University of Perugia**  
One academic year of chemistry courses and laboratory work in an international environment.

## SELECTED PUBLICATIONS

## FIRST AUTHOR · JACS · 2026

Adsorption and Sulfur-Selective Photooxidation of Cysteine on Anatase TiO<sub>2</sub>(101). 148(25), 26710–26723. DOI: 10.1021/jacs.6c07370

## FIRST AUTHOR · JACS · 2025

Unraveling the Role of the Multifunctional Groups in the Adsorption of L-Cysteine on Rutile TiO<sub>2</sub>(110). 147(44), 40158–40170. DOI: 10.1021/jacs.5c07119

## SECOND AUTHOR · JPCC · 2026

In-Situ Growth and Oxidation of Cu Nanoparticles on Rutile TiO<sub>2</sub>(110). 130(20), 7136–7147. DOI: 10.1021/acs.jpcc.6c00835

## CO-AUTHOR · GISAXS · ACS AMI · 2024

Light-Induced Transformation of Virus-Like Particles on TiO<sub>2</sub>. 16(28), 37275–37287. DOI: 10.1021/acsami.4c07151

## CAPABILITIES

## MATERIALS &amp; METHODS

Surface and interface science · TiO<sub>2</sub> and ZnO · Thin films · Photocatalysis · Energy materials · Recycled polymers · STM/SPM · XPS · FT-IRRAS · AFM · SEM · TEM · XRD · LEED · GISAXS · GIXRD · UHV · Electrochemistry

## PROJECTS &amp; DIGITAL

Work-package coordination · Experimental strategy · Stakeholder collaboration · PM2 · Technical reporting · Scientific writing · Python · TypeScript · Node.js · LLM applications · Google ADK · Model Context Protocol · Multi-agent workflows